Domain and Range

ANSWER KEY

Section 1: Finding Domains of Rational and Radical Functions

1 Find the domain of $f(x) = \frac{x-3}{x^2-16}$.

a. $(-\infty, -4) \cup (-4, 4) \cup (4, \infty)$ ✓

b. $(-\infty, 4) \cup (4, \infty)$

c. $[-4, 4]$

d. All real numbers except $x = 3$

Denominator cannot be zero: $x^2 - 16 \neq 0 \Rightarrow x \neq \pm 4$. Domain: all reals except $x = -4$ and $x = 4$.

2 Find the domain of $g(x) = \sqrt{2x-8}$.

a. $[4, \infty)$ ✓

b. $(4, \infty)$

c. $[8, \infty)$

d. $(-\infty, 4]$

The radicand must be non-negative: $2x - 8 \geq 0 \Rightarrow 2x \geq 8 \Rightarrow x \geq 4$, which is $[4, \infty)$.

3 Find the domain of $h(x) = \frac{1}{\sqrt{9-x^2}}$.

a. $(-3, 3)$ ✓

b. $[-3, 3]$

c. $(-\infty, -3) \cup (3, \infty)$

d. $[0, 3)$

Radicand in denominator must be strictly positive: $9 - x^2 > 0 \Rightarrow x^2 < 9 \Rightarrow -3 < x < 3$.

Section 2: Finding Ranges of Algebraic Functions

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4 Find the range of $f(x) = x^2 - 4x + 7$.

a. $[3, \infty)$ ✓

b. $[7, \infty)$

c. $(-\infty, 3]$

d. $[0, \infty)$

Complete the square: $f(x) = (x - 2)^2 + 3$. Since $(x - 2)^2 \geq 0$, the minimum value is 3. Range: $[3, \infty)$.

5 Find the range of $g(x) = -|x + 2| + 5$.

a. $(-\infty, 5]$ ✓

b. $[5, \infty)$

c. $(-\infty, -2]$

d. $[-5, 5]$

Since $|x + 2| \geq 0$, $-|x + 2| \leq 0$. Adding 5 gives $g(x) \leq 5$. Range: $(-\infty, 5]$.

6 Find the range of $h(x) = \sqrt{25 - x^2}$.

a. $[0, 5]$ ✓

b. $[-5, 5]$

c. $[0, 25]$

d. $[5, \infty)$

Domain is $[-5, 5]$. Min occurs at $x = \pm 5$ ($h = 0$). Max occurs at $x = 0$ ($h = 5$). Range is $[0, 5]$.

Section 3: Piecewise Functions

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7 Evaluate $f(-2)$ for $f(x)$ defined as $2x+1$ for $x<0$, and x^2-3 for $x \geq 0$.

a. -3 ✓

b. 1

c. -2

d. 7

Since $-2<0$, use the top rule: $f(-2)=2(-2)+1=-4+1=-3$.

8 Find the domain of $f(x)$ defined as $\frac{1}{x+3}$ for $x<1$, and $\sqrt{x-1}$ for $x \geq 1$.

a. $(-\infty, -3) \cup (-3, \infty)$ ✓

b. $[1, \infty)$

c. All real numbers

d. $(-\infty, 1) \cup [1, \infty)$

For $x<1$, $x+3 \neq 0 \Rightarrow x \neq -3$. For $x \geq 1$, $x-1 \geq 0$ is always true. Domain is all reals except $x=-3$.

9 Determine the range of $g(x)$ defined as $-x$ for $x<0$, and $x+1$ for $x \geq 0$.

a. $(0, \infty)$ ✓

b. $[1, \infty)$

c. $[0, \infty)$

d. All real numbers

For $x<0$, $-x>0$, giving outputs $(0, \infty)$. For $x \geq 0$, $x+1 \geq 1$, giving $[1, \infty)$. Union is $(0, \infty)$.

Section 4: Domain and Range of Compositions

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10 If $f(x) = \frac{1}{x-2}$ and $g(x) = \sqrt{x}$, find the domain of $(f \circ g)(x)$.

a. $[0, 4) \cup (4, \infty)$ ✓

b. $[0, \infty)$

c. $(4, \infty)$

d. All reals except $x = 2$

First, $g(x)$ requires $x \geq 0$. Second, $f(g(x)) = \frac{1}{\sqrt{x}-2}$ requires $\sqrt{x} \neq 2 \Rightarrow x \neq 4$. Domain: $[0, 4) \cup (4, \infty)$.

11 If $f(x) = x^2$ and $g(x) = 3x - 1$, what is the range of $(f \circ g)(x)$ for $x \in \mathbb{R}$?

a. $[0, \infty)$ ✓

b. $[-1, \infty)$

c. $[1, \infty)$

d. All real numbers

$(f \circ g)(x) = (3x - 1)^2$. Any real squared is non-negative, so the range is $[0, \infty)$.

12 Find the domain of $(g \circ f)(x)$ if $f(x) = 2x + 6$ and $g(x) = \ln(x)$.

a. $(-3, \infty)$ ✓

b. $[0, \infty)$

c. $(-6, \infty)$

d. $(3, \infty)$

$(g \circ f)(x) = \ln(2x + 6)$. The argument of logarithm must be strictly positive:
$2x + 6 > 0 \Rightarrow 2x > -6 \Rightarrow x > -3$.

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